From Classroom/Integrated School Programs 7 in Top 20, 23 in Top 100, 54 in Top 300, 106 in Top 500 All India Ranks & 2314 Students from Classroom /Integrated School Programs & 3723 Students from All Programs have been Awarded a Rank in JEE (Advanced), 2013

ALL INDIA TEST SERIES

ANSWERS, HINTS & SOLUTIONS

FULL TEST – II

PAPER-1

Q.NO	PHYSICS	CHEMISTRY	MATHEMATICS
1.	A	A	C
2.	C	A	D
3.	B	A	A
4.	C	A	B
5.	C	A	B
6.	A	A	C
7.	A, B, C, D	A, B, C, D	A, C, D
8.	A, B	A, C, D	A, C
9.	C, D	A, B, C	A, B
10.	B, C	A, D	A, D
11.	B	B	C
12.	B	A	B
13.	B	A	A
14.	C	C	A
15.	A	B	C
16.	B	C	D
17.	A	C	C
18.	A	B	B
19.	C	D	C
20.	D	C	D
21.	C	D	B
22.	D	B	D
23.	C	A	A

Physics

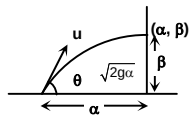
PART – I

2. $(v_1\cos\alpha)t = (v_2\cos\beta)t = 30\sqrt{3}$ . . . (1)

$(v_1\sin\alpha)t - \frac{1}{2}gt^2 + h = (v_2\sin\beta)t - \frac{1}{2}gt^2$ . . . (2)

from (1) and (2)

h = 60 m

3. 

$\beta = \alpha\tan\theta - \frac{1}{2}g\frac{\alpha^2}{u^2\cos^2\theta}$

$\beta = \alpha\tan\theta - \frac{1}{2}g\frac{\alpha^2}{2g\alpha}\left(1+\tan^2\theta\right)$

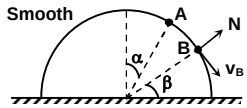
$\tan^2\theta - 4\tan\theta + \left(1+\frac{4\beta}{\alpha}\right) = 0$

If particle will not hit the target.

$(b^2 - 4ac) < 0$

$16 - 4\left(1+\frac{4\beta}{\alpha}\right) < 0$

$4\beta > 3\alpha$

4. 

At B, $mg\sin\beta = \frac{mv_B^2}{r}$ . . . (1)

Using energy considerations

$\frac{1}{2}mv_B^2 = mgr(\cos\alpha - \sin\beta)$ . . . (2)

From (1) and (2)

$mg\sin\beta = 2mg(\cos\alpha - \sin\beta)$

$2\cos\alpha = 3\sin\beta$

7. Here the position y on the screen will correspond to maxima.

$y = \frac{n\lambda D}{d}$

$\lambda = \frac{30{,}000}{n}$ when n = 1, 2, 3, 4, ……..

8. $a = kx^2$

$\frac{da}{dx} = 2kx \Rightarrow \tan 60° = 2k\sqrt{3} \Rightarrow k = \frac{1}{2}$

$a = \frac{x^2}{2}$

$v\frac{dv}{dx} = \frac{x^2}{2} \Rightarrow \frac{v^2}{2} = \frac{x^3}{6} + C_1$ at x = 0, v = 3 m/s

$C_1 = \frac{9}{2}$

$v^2 = \frac{x^3}{3} + 9$

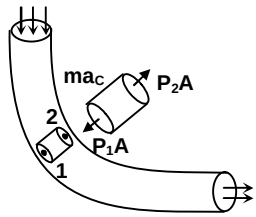
hence a = 1.5 and $v = \sqrt{\sqrt{3}+9}$

9. Capillary height $h = \dfrac{2T\cos\theta}{r\rho g} \Rightarrow h = 59.6$ mm

here 59.6 mm is greater than the protruding part hence water will rise in the capillary of insufficient height 25 mm.

Now, $R = \dfrac{2T}{hrg} = 0.6$ mm

10. 10. Fluid particles passing through the bend are in circular motion. The centripetal acceleration is provided by the variation in pressure. In the section as shown



$P_1A - P_2A = ma_C$

$\Rightarrow P_1 > P_2$

applying Bernoulli theorem at 1 and 2

i.e., $P_1 + \dfrac{1}{2}\rho_1 v_1^2 + \rho g h_1 = P_2 + \dfrac{1}{2}\rho v_2^2 + \rho g h_2$

$P_1 + \dfrac{1}{2}\rho_1 v_1^2 = P_2 + \dfrac{1}{2}\rho v_2^2$ as $h_1 = h_2$

$\because P_1 > P_2$

$v_1 < v_2$

15. $m_{Ag}\sin\theta = 5 \times 10 \times \sin 37° = 5 \times 10 \times \dfrac{3}{5} = 30$ N

$f_{max} = \mu_{AB}(m_C + m_B)g\cos\theta = 48$ N

hence f = 30 N

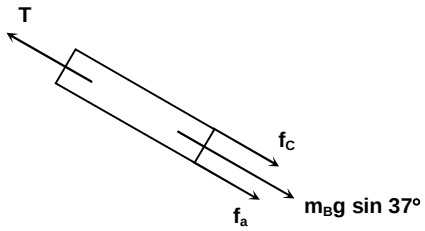
16. $m_C g\sin\theta = 2\times10\times\dfrac{3}{5} = 12$N

$f_{max} = \mu m_c g\cos\theta = 0.1\times2\times10\times\dfrac{4}{5} = 1.6$ N

hence f = 1.6 N

17. $T = f_A + f_C + m_B g\sin37°$

= 91.6 N



18. Optical path difference between the beans arriving at P

$\Delta x = (\ell_2 - \ell_1) + d\sin\theta$

for maxima

$(\ell_2 - \ell_1) + d\sin\theta = n\lambda$

$\sin\theta = \dfrac{1}{d}[n\lambda - (\ell_2 - \ell_1)]$

$\theta = \sin^{-1}\left[2\left(\dfrac{n}{40} - 1\right)\right]$

19. $|\sin\theta| \le 1$

$$-1 \le 2\left[\frac{n}{40} - 1\right] \le 1$$

or $20 \le n \le 60$

Hence number of maxima = 60 – 20 = 40

20. At C, phase difference

$$\phi = \frac{2\pi}{\alpha}(\ell_2 - \ell_1) = 80\ \pi$$

Hence for maximum intensity will appear at C.

Now for minimum intensity at C

$$(\mu - 1)t = \frac{\lambda}{2}$$

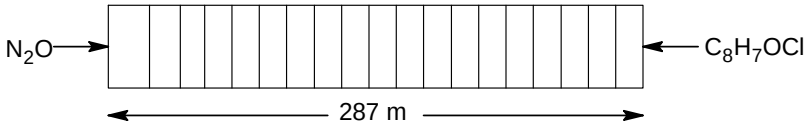
$$t = \frac{t}{2(\mu - 1)} = 500\ \text{nm}$$

Chemistry

PART – II

3. Neither N nor F contain d-orbitals. Further in N_2F_4 N—N bond is shorter than in N_2H_4 due to s-character (Bent's Rule).

4. Due to poor shielding effect of 3d electron the size of Br^- ion nearly same as that of Cl^- ion.

5. 

Molar mass of $C_8H_7OCl = 8 \times 12 + 7 \times 1 + 16 + 35.5 = 154.5$ g
Molar mass of $N_2O = 2 \times 14 + 16 = 44$g
According to Graham's Law of diffusion

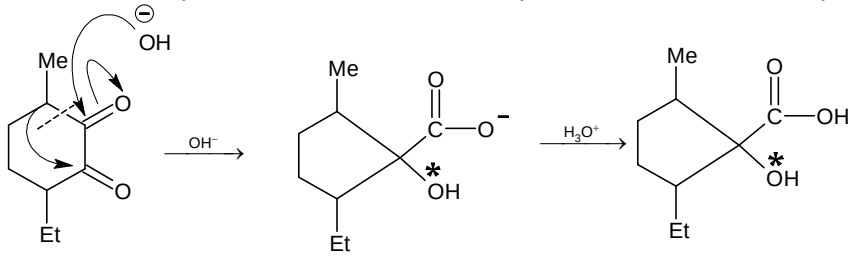
$$\frac{r_{N_2O}}{r_{C_8H_7OCl}} = \sqrt{\frac{M_{C_8H_7OCl}}{M_{N_2O}}} = \sqrt{\frac{154.5}{44}} = \sqrt{3.5} = 1.87 : 1$$

$$d_{N_2O} = \frac{1.87}{2.87} \times 287 = 187^{th} \text{ row from } N_2O \text{ side}$$

$$d_{C_8H_7OCl} = \frac{1.0}{2.87} \times 287 = 100^{th} \text{ row from weeping gas side}$$

Therefore, the spectator from the side of N_2O in the 187th row will be laughing and weeping simultaneously Alternatively, the spectator from the side of weeping gas in 100th row will laugh and weep.

6. OH^- attacks at the more reactive (C=O) group, (containing more EWG or less EDG) Et is more EDG than Me (due to +I effect here no H.C.). Therefore OH^- attack (C=O) with (Me) group.

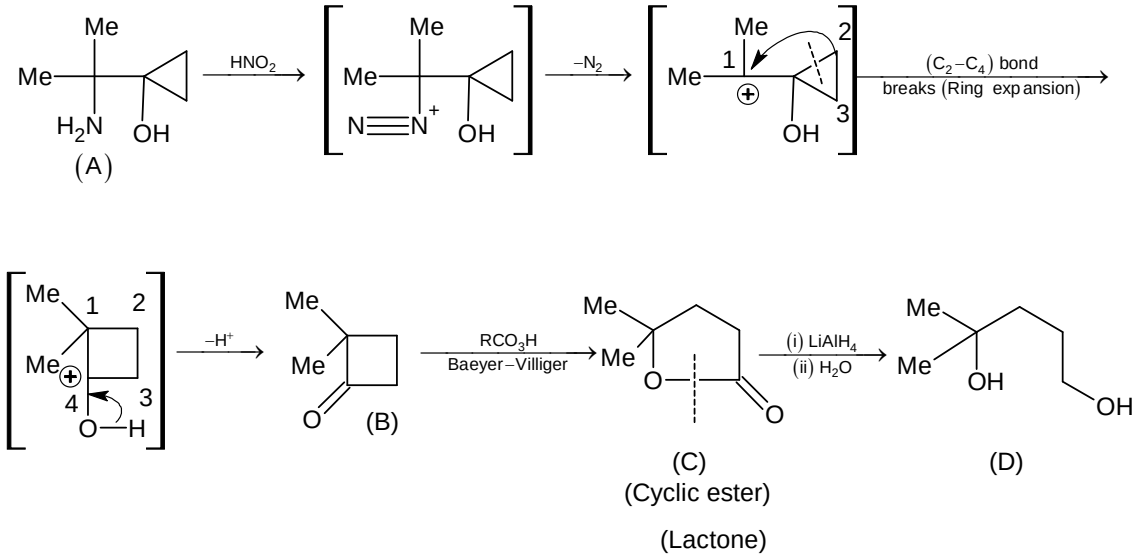


10. $NH_4NO_3\,(aq) \longrightarrow N_2O + H_2O$
$NH_2OH + HNO_2 \longrightarrow N_2O + 2H_2O$
Hence, (A) and (D) are the correct answers.

13. Velocity = $2.18 \times 10^6 \times \frac{Z}{n}$

14. (i) $E_{BrO_3^-} = M/5$
(ii) $E_{BrO_3^-} = M/6$
$\therefore$ ratio = 6/5

17.



18. $\underset{0.8\times15\atop=12}{2KIO_3} + \underset{35\times0.15\atop=5.25}{Pb(NO_3)_2} \rightleftharpoons Pb(IO_3)_2 + 2KNO_3$

1 mol of $Pb(NO_3)$ reacts with 2 mol of KIO_3
5.25 m mol of $Pb(NO_3)_2$ reacts = 2×5.25
= 10.5 m mol
m mol of KIO_3 left = 12 – 10.5
= 1.5
Hence $Pb(NO_3)_2$ is the limiting reagent

19. $\left[IO_3^-\right] = \dfrac{1.5 \text{ m mol}}{(15+35)\text{ mL}} = 0.03 \text{ M}$

20. Due to common ion $\left(IO_3^-\right)$ left in the solution the solubility $Pb(IO_3)_2$ decreases.

$S = \dfrac{K_{sp}}{(0.03)^2} = \dfrac{2.7\times10^{-13}\times10^4}{9}$

$= 0.30 \times 10^{-9}$
$= 3.0 \times 10^{-10}$

21. pH range of all these indicators lies in between 2-12. Hence, all are suitable indicators.

22. Since curve is given for strong acid and strong base hence, pH at the end point is 7.

Mathematics

PART – III

1. $y = ax^2 + bx + c$, vertex is (4, 2)

$\Rightarrow 4 = -\frac{b}{2a}$, $b = -8a$, $\frac{4ac - b^2}{4a} = 2 \Rightarrow c = 2 + \frac{b^2}{4a} = 2 + 16a$

Now $\Delta = abc = -8a^2(2 + 16a) = -16(a^2 + 8a^3)$

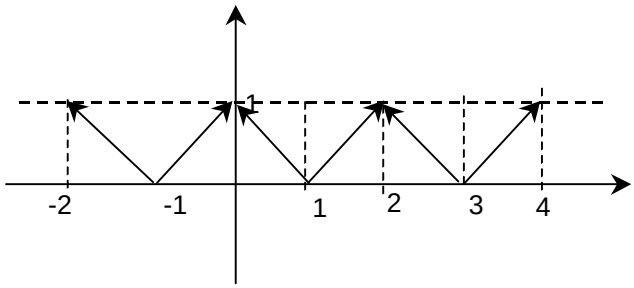
$\Rightarrow \frac{d\Delta}{da} = -16(2a + 24a^2) < 0 \ \forall \ a \in [1, 3]$

$\Rightarrow \Delta|_{max} = -144$, $\Delta|_{min} = -3600 \Rightarrow$ Difference = 3456

2. $x - [x] = \{x\}$

$x - [x + 1] = \{x\} - 1$

$\int_{-2}^{4} f(x)dx = 6.\frac{1}{2}(1.1) = 3$



3. $\lim_{x\to 0} \int_0^x \frac{t^2dt}{(x - \sin x)\sqrt{a + t}} = \lim_{x\to 0} \frac{\int_0^x \frac{t^2dt}{\sqrt{a+t}}}{(x - \sin x)} = \lim_{x\to 0} \frac{x^2}{\sqrt{a + x}\,(1 - \cos x)}$

$= \lim_{x\to 0} \frac{x^2}{2\sin^2\frac{x}{2}.\sqrt{a + x}} = \frac{2}{\sqrt{a}} = 1$ (given)

$\Rightarrow a = 4$

4. $(3)^6 = 729 < 900$ and $(3)^7 = 2187 > 900$

5. Now by property of triangles

$RQ = \frac{1}{2}BC = \frac{a}{2}$

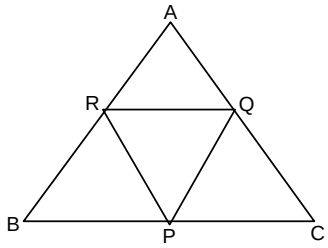
Similarly $PQ = \frac{c}{2}$, $PR = \frac{b}{2}$

Area of $\Delta ABC = \frac{abc}{4R}$

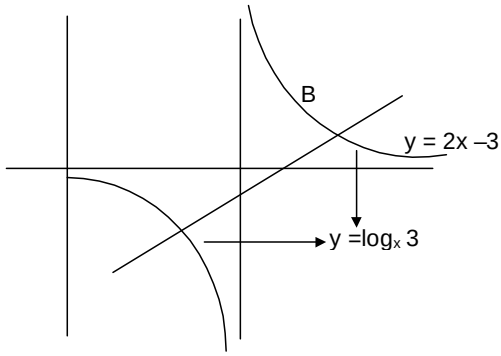
That of $\Delta PQR = \frac{\frac{a}{2}\frac{b}{2}\frac{c}{2}}{4R'}$

Also area of ΔABC = 4 (area of ΔPQR)

$\Rightarrow \frac{abc}{32R'} = \frac{abc}{16R} = \left(\frac{R}{R'} = \frac{2}{1}\right)$

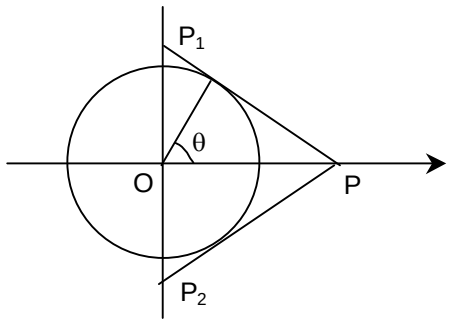


6. Clearly 2 solutions



7. $f(x) = \cos \pi(|x| + [x])$
$f(0) = 1$, $f(0 - 0) = \cos \pi(-1) = -1$
$f(1/2) = \cos \pi\left(\frac{1}{2}\right) = 0$, $f\left(\frac{1}{2} - 0\right) = \cos \pi\left(\frac{1}{2}\right) = 0$
$f\left(\frac{1}{2} + 0\right) = \cos\frac{\pi}{2} = 0$
for $x \in (0, 1)$ $f(x) = \cos \pi x$
for $x \in (-1, 0)$ $f(x) = \cos \pi (-x - 1) = -\cos \pi x$

8. $OP = 5\sqrt{2}\ \sec\theta$
$OP_1 = 5\sqrt{2}\ \text{cosec}\ \theta$
$\Delta PP_1P_2 = \frac{100}{\sin 2\theta}$
$(\Delta \mid PP_1P_2)_{min} = 100$
$\Rightarrow \theta = \pi/4$
$\Rightarrow OP = 10$
$\Rightarrow P = (10, 0), (-10, 0)$



9. Let P(2, –1) goes 2 units along $x + y = 1$ upto A and 5 units along $x - 2y = 4$ up to B
Slope of PA = –1 = tan 135°, slope of PB = $\frac{1}{2}$ = tan θ
$\therefore \sin\theta = \frac{1}{\sqrt{5}}$, $\cos\theta = \frac{2}{\sqrt{5}}$
$\therefore A \equiv (x_1 + r\cos 135°, y_1 = r \sin 135°) = (2 + 2x\ -\frac{1}{\sqrt{2}}, -1 + 2 \times \frac{1}{\sqrt{2}}) = (2 - \sqrt{2}, \sqrt{2} - 1)$
$B \equiv (x_1 + r\cos\theta, y_1 + r\sin\theta) = (2 + 5 \times \frac{2}{\sqrt{5}}, -1 + \frac{5}{\sqrt{5}}) = (2\sqrt{5} + 2, \sqrt{5} - 1)$

10. Slope of tangent = $\frac{\frac{-1}{a}}{\frac{1}{b}} = -\frac{b}{a}$

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{\frac{-4}{t^2}}{4} = -\frac{1}{t^2} < 0$$

So, $\frac{-b}{a} < 0 \Rightarrow \frac{b}{a} > 0$

12. $P\left(\frac{\bar{A}}{\bar{B}}\right) = \frac{1-P(A \cup B)}{P(\bar{B})} = \frac{1-x}{x^2} \le 1$

$\Rightarrow x^2 + x - 1 \ge 0$

$\Rightarrow x \ge \frac{\sqrt{5}-1}{2}$ or $x \le \frac{-1-\sqrt{5}}{2}$

$\therefore$ x is positive

$x \ge \frac{\sqrt{5}-1}{2}$

13. $\left(\frac{a+b+c+d}{4}\right)^2 \le \frac{a^2+b^2+c^2+d^2}{4}$

$\frac{(8-e)^2}{16} \le \frac{16-e^2}{4}$

$64 + e^2 - 16e \le 64 - 4e^2$

$5e^2 - 16e \le 0$

$0 \le e \le \frac{16}{5}$

14. Let A(a, P(a)), B(b, P(b)), then slope of AB = P′(a) = P′(b) from LMVT $\exists$ c $\in$ (a, b)
Where P′(c) = slope of AB

15. Given QT = QA = 1

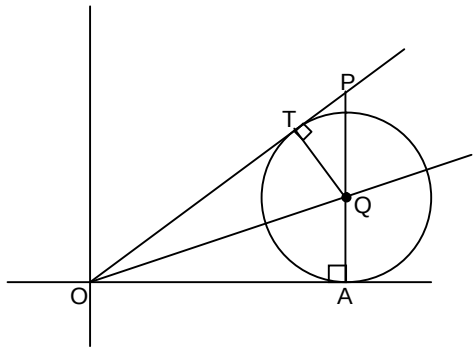
Let PQ = x, then $PT = \sqrt{x^2-1}$

Then Δ TQP and Δ APO are similar triangles

Then $OT = OA = \frac{x+1}{\sqrt{x^2-1}}$

$\Rightarrow 1 + x + \frac{2(x+1)}{\sqrt{x^2-1}} + \sqrt{x^2-1} = 8$

$\Rightarrow x = \frac{5}{3}$



16. From above, OA = 2 and AQ = 1
$\Rightarrow$ coordinate of Q $\equiv$ (2, 1)
$\Rightarrow$ equation is $(x-2)^2 + (y-1)^2 = 1$

17. ΔORQ is a right angled triangle. Then O and R are the extremities of the diameter then the coordinates of R($2\sqrt{3}$, 2)

$\Rightarrow$ Equation of circle $(x-0)(x-2\sqrt{3}) + (y-0)(y-2) = 0 \Rightarrow x^2 + y^2 - 2\sqrt{3}x - 2y = 0$

18. Perpendicular tangents intersect on the director circle of hyperbola and director circle of rectangular hyperbola is a point circle. Hence centre of hyperbola is (1, 1) and equation of asymptotes are $(x - 1) = 0$ and $y - 1 = 0$

19. Equation of hyperbola is $xy - x - y + 1 + \lambda = 0$
It passes through (3, 2) hence $\lambda = -2$
Equation of hyperbola is $xy = x + y + 1$

20. From the centre of hyperbola we can draw two real normals to the rectangular hyperbola

21. Taking point $(r\cos\theta, r\sin\theta)$, we get

$$r^2 = \frac{3+\sqrt{2}}{3-\sqrt{2}\sin\left(2\theta+\frac{\pi}{4}\right)}$$

$r_{max} = \sqrt{\frac{3+\sqrt{2}}{3-\sqrt{2}}}$, $r_{min} = 1$

Max exist when $2\theta + \frac{\pi}{4} = \frac{\pi}{2} \Rightarrow \theta = \frac{\pi}{8}$

22. Required equation is $\frac{x^2}{\frac{3+\sqrt{2}}{3-\sqrt{2}}} + y^2 = 1$

23. Centre of circle be origin and its radius is the length of semi minor axis
Hence area = π